

JS Looping Statements

1. What are Looping Statements?

- a. A looping statement is a way to repeat a block of code multiple times based on a condition.
- b. This helps in doing repetitive tasks.
- c. Instead of writing the same code over and over again, loops make the process efficient and easy to understand.

2. Why Do We Need Looping Statements in JS?

- a. Looping is essential because it allows us to:
 - i. Perform repetitive tasks without repeating code.
 - ii. Save time and reduce errors.
 - iii. Handle tasks like iterating through arrays, creating patterns, or processing data in chunks.

3. Types of Looping Statements:

- a. JavaScript has several types of loops, each suited for different scenarios:
 - i. For Loop
 - ii. While Loop
 - iii. Do-While Loop

4. For-Loop:

- a. A **for** loop is used when you know how many times you want the code to run. It includes three parts:
 - i. **Initialization**: Setting up a counter.
 - ii. **Condition**: A check to continue looping.
 - iii. **Increment / Decrement**: Update the counter.

b. **Syntax** →

```
💡 for (initialization; condition; increment/decrement) {  
    // Code to execute  
}
```

5. While Loop:

- a. A **while** loop runs as long as a condition is true.
- b. If the condition is false from the start, it won't run.
- c. **Syntax** →

```
💡 while (condition) {  
    // Code to execute  
    //updatation  
}
```

6. Do-While Loop:

- a. A **do-while** loop is similar to the **while** loop but it runs at least once, because the condition is checked after the code executes.
- b. **Syntax** →

```
💡 do {  
    // Code to execute  
    //updatation  
}  
while (condition);
```

7. Break and Continue:

- a. **break**: Used to exit the loop immediately, no matter where it is in the iteration.
- b. **continue**: Skips the current iteration and moves to the next one.
- c. **Example of Break:**

```
1 for (let i = 0; i < 5; i++) {  
2   if (i === 3) {  
3     break; // Exit the loop when i is 3  
4   }  
5   console.log(i); // Output: 0, 1, 2  
6 }
```

- d. **Example of Continue:**

```
1 for (let i = 0; i < 5; i++) {  
2   if (i === 3) {  
3     continue; // Skip the iteration when i is 3  
4   }  
5   console.log(i); // Output: 0, 1, 2, 4  
6 }
```